Assignment III: AI For Engineers

November 2025

PART 1: Structural Questions

Q1. Probabilistic Reasoning and Bayes' Theorem

a. Compute $P(\textbf{Spam} \mid \textbf{Feature Present})$ using Bayes' theorem.

- Prior Probability of Spam: $P(\text{Spam}) = 0.3$
- Prior Probability of Not Spam: $P(\text{Not Spam}) = 1 - 0.3 = 0.7$
- Likelihood of Feature given Spam: $P(\text{Feature} \mid \text{Spam}) = 0.8$
- Likelihood of Feature given Not Spam: $P(\text{Feature} \mid \text{Not Spam}) = 0.1$

The goal is to compute the posterior probability $P(\text{Spam} \mid \text{Feature})$ using Bayes' Theorem:

$$P(\text{Spam} \mid \text{Feature}) = \frac{P(\text{Feature} \mid \text{Spam})P(\text{Spam})}{P(\text{Feature})}$$

Step 1: Calculate the Marginal Probability of the Feature, $P(F)$
Using the Law of Total Probability for mutually exclusive events:

$$P(F) = P(F \mid S)P(S) + P(F \mid \neg S)P(\neg S)$$

$$P(F) = (0.8 \times 0.3) + (0.1 \times 0.7)$$
$$P(F) = 0.24 + 0.07 = 0.31$$

Step 2: Apply Bayes' Theorem

$$P(S \mid F) = \frac{P(F \mid S)P(S)}{P(F)}$$

$$P(S \mid F) = \frac{0.8 \times 0.3}{0.31} = \frac{0.24}{0.31}$$
$$P(S \mid F) \approx 0.7742$$

b. Interpret this probability means in practical terms.

The calculated probability, $P(\text{Spam} \mid \text{Feature Present}) \approx 0.7742$, represents the posterior probability that an email belongs to the spam class after observing that the specific feature is present.

In practical terms, it means that the presence of this single feature raises the model's belief that the email is spam from the initial prior probability of 30% to approximately 77.42%. This indicates that the feature is a strong diagnostic indicator for spam.

c. Now suppose the model uses two independent features, $F1$ and $F2$. Describe how the Naïve Bayes assumption simplifies the computation of $P(\textbf{Spam} \mid F1, F2)$.

The general formulation using Bayes' Theorem to compute the posterior probability with two features is:

$$P(S \mid F1, F2) = \frac{P(F1, F2 \mid S)P(S)}{P(F1, F2)}$$

The Naïve Bayes assumption drastically simplifies this computation by assuming that the two features, $F1$ and $F2$, are conditionally **independent** of one another given the class variable (Spam/Not Spam).

Under this assumption, the complex joint likelihood term, $P(F1, F2 \mid S)$, can be factored into a product of individual conditional probabilities:

$$P(F1, F2 \mid S) = P(F1 \mid S) \times P(F2 \mid S)$$

This simplification allows the learning problem to decompose into separate, smaller problems, where the training only requires determining the probability of each feature given the class, $P(F \mid S)$, instead of complex joint probabilities. This makes the learning process extremely fast and scalable.

d. Briefly explain one advantage and one limitation of the Naïve Bayes model.

- **Advantage:** The Naïve Bayes algorithm is extremely fast and highly scalable for large datasets, particularly those with many features (e.g., text data). Because the learning task decomposes into independent parameter estimation problems due to the conditional independence assumption, the model training is not iterative and the time required is linear with the number of features.
- **Limitation:** The core assumption of conditional independence is often violated in real-world applications.For Example: words in a sentence are not truly independent. This violation causes the model to produce overconfident probability estimates that are often numerically skewed too close to 0 or 1. While the final categorical classification is often accurate, the numerical probability assigned to that classification can be unreliable.

Q2. Decision Processes and Expected Utility

a. Compute the expected utility (EU) for each route assuming:

$$\text{Utility} = (\text{Probability} \times \text{Reward}) - \text{Cost}$$

Route A:

$$EU(A) = (0.9 \times 120) - 20 = 108 - 20 = 88$$

Route B:

$$EU(B) = (0.7 \times 150) - 10 = 105 - 10 = 95$$

Route C:

$$EU(C) = (0.5 \times 200) - 5 = 100 - 5 = 95$$

b. Which route should the drone choose under the maximum expected utility principle? (Use the route with the least Cost as a tie-breaker if there are any ties)

- **Maximum Expected Utility:** Routes B and C both yield the maximum expected utility of 95.
- **Tie-breaker (Least Cost):**
  - $Cost(B) = 10$
  - $Cost(C) = 5$

The drone should choose Route C because it has the same maximum expected utility as Route B (95) but has a lower associated cost (5 vs. 10).

c. Discuss briefly how Markov Decision Processes (MDPs) extend this concept to sequential decision-making.

Markov Decision Processes (MDPs) extend the concept of Bayesian Networks to sequential decision-making under uncertainty by introducing actions and rewards over time. While Bayesian Networks model probabilistic dependencies between variables at a single point, MDPs model a system that evolves through discrete time steps. An MDP is defined by a tuple (S, A, P, R), where S is a set of states, A is a set of actions, $P(s' \mid s, a)$ is the transition probability of moving from state s to state s' after taking action a, and $R(s, a)$ is the reward received for taking action a in state s.

The key idea is that the next state depends only on the current state and action (the Markov property), allowing agents to plan sequences of actions that maximize expected cumulative reward. In this way, MDPs generalize probabilistic reasoning to dynamic environments, enabling decision-making over time rather than at a single snapshot.

Q3. Probabilistic Graphical Models

a. Define what a Bayesian Network is and explain how it represents dependencies between variables.

A Bayesian Network is a probabilistic model that uses a directed acyclic graph to show how different variables in a system depend on one another. Each variable is represented as a node, and arrows between nodes show direct influences, while conditional probability tables describe how likely each variable is given the ones that affect it. For example, Weather might influence whether a Sprinkler is turned on or whether it Rains, and both Sprinkler and Rain together determine whether the Grass becomes Wet. This relationship can be expressed mathematically as

$$P(W, S, R, G) = P(W) \cdot P(S \mid W) \cdot P(R \mid W) \cdot P(G \mid S, R)$$

meaning that each variable only depends on its parents in the graph. In this way, a Bayesian Network simplifies complex systems by clearly showing which variables depend on which others and allowing us to reason efficiently about uncertainty.

The semantics of the Bayes Net defines the full joint probability distribution $P(X_1, ..., X_n)$ as the product of the local conditional distributions, as given by the chain rule for Bayesian networks:

$$P(X_1, ..., X_n) = \prod_{i=1}^{n} P(X_i \mid \text{Parents}(X_i))$$

This factorization allows a Bayes Net to be exponentially more compact than enumerating the full joint distribution, provided the dependency structure is sparse (i.e., each variable has a small number of parents).

B.Draw a simple Bayesian Network for a medical diagnosis system with variables: Disease, Test Result, and Symptoms. Label dependencies clearly.

A simple Bayesian Network for a medical diagnosis system can be represented with the variable Disease influencing both Symptoms and Test Result. This shows that the presence of a disease affects what symptoms appear and how the diagnostic test result turns out.

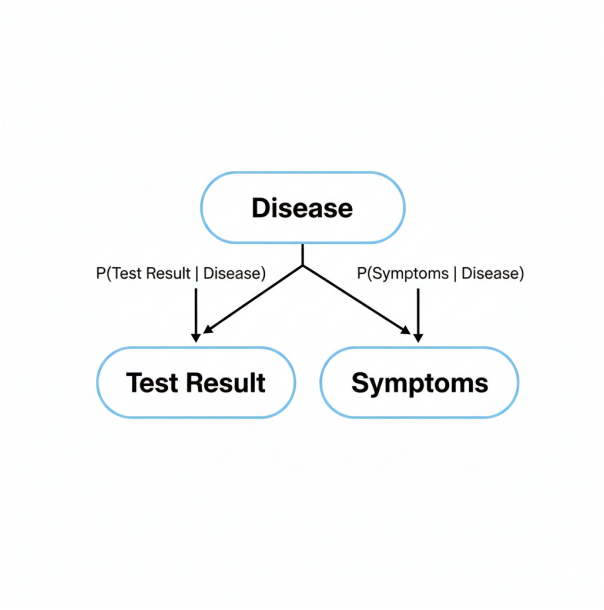


Figure 1: Bayesian Network for Medical Diagnosis

$$\text{Disease} \rightarrow \text{Symptoms}$$

$$\text{Disease} \rightarrow \text{Test Result}$$

This structure indicates that both the symptoms experienced by a patient and the test results depend directly on the underlying disease.

c. Explain one advantage of using Bayesian Networks for reasoning under uncertainty.

One key advantage of using Bayesian Networks for reasoning under uncertainty is that they allow efficient probabilistic inference by explicitly modeling dependencies between variables. Because a Bayesian Network represents the joint probability distribution as a product of conditional probabilities, it reduces the complexity of computing probabilities in large systems. Instead of considering all variables simultaneously, the network only considers a variable's direct parents when calculating probabilities. This not only makes reasoning more computationally feasible but also helps in understanding which factors directly influence outcomes, enabling more accurate predictions and decision-making under uncertainty.

Q4. Supervised Learning: Classification Model

4.5 Discuss which model performed better and provide a brief explanation based on the data characteristics (e.g., feature types, linearity, interpretability)

When we compare the performance of the two models, the Logistic Regression model edges out the Decision Tree classifier overall. Looking at the numbers, Logistic Regression achieved a slightly higher accuracy (0.8025 vs. 0.7866) and a noticeably higher recall (0.7206 vs. 0.6324). This tells us that it was better at correctly identifying passengers who actually survived.

On the other hand, the Decision Tree did show a slightly higher precision (0.8350 vs. 0.8033), which means that when it predicted a passenger would survive, it was a bit more likely to be correct. However, its lower recall indicates that it missed more of the true survivors.

These differences make sense when we consider the nature of the dataset. The features we relied on passenger class, sex, and age tend to relate to survival in a mostly linear way. Logistic Regression, being a model that captures linear trends efficiently, handles this well and generalizes nicely to new, unseen data. The Decision Tree, by contrast, tries to draw complex boundaries even from just three predictors, which can lead it to overfit.

Q5. Unsupervised Learning: Clustering Analysis

5.3 Compute the Silhouette Score for your clustering results and interpret what the score indicates about the quality of your clusters.

The Silhouette Score we obtained is 0.3945, which suggests a moderate level of clustering quality. The Silhouette Score ranges from 1 to +1 and reflects how well-separated and distinct the clusters are. A value around 0.39 indicates that the K-Means algorithm did uncover some meaningful groupings within the passenger data, but the clusters are not sharply defined and show noticeable overlap.

This overlap is not surprising, given that the clustering was based on only three features: passenger class, sex, and age. In reality, survival on the Titanic was influenced by many additional factors—such as fare paid, family size, cabin location, and proximity to lifeboats that were not included in the model. As a result, passengers with similar demographics might have ended up in the same cluster but experienced very different outcomes. Therefore, while the clusters capture some underlying patterns, the moderate Silhouette Score reflects the complexity and nuance of the actual survival data.

5.4 Compare the cluster groupings with the actual survival labels (survived) to discuss any patterns or similarities.

The cluster assigns in relation to the actual survival outcomes, some intriguing patterns begin to emerge. Cluster 0, for instance, contains 523 non-survivors and 135 survivors, indicating that the majority of passengers in this group did not survive. This suggests that this cluster may be capturing characteristics associated with higher mortality, such as being older, male, or traveling in lower-class accommodations. In contrast, Cluster 1 is composed of 96 non-survivors and 292 survivors, meaning that most passengers in this group did survive. This pattern implies that Cluster 1 tends to group passengers with traits linked to a higher likelihood of survival, such as being younger, female, or in higher-class cabins. In essence, the clustering appears to roughly divide the passengers into higher-risk (Cluster 0) and lower-risk (Cluster 1) categories. However, the separation is not perfect,there are still survivors within Cluster 0 and non-survivors within Cluster 1 reflecting the moderate Silhouette Score we noted earlier. Overall, these clusters do capture meaningful structure within the data, highlighting the relevance of features like age, sex, and class, while also reminding us that survival outcomes are influenced by additional factors not fully captured by this clustering approach.

5.5 Briefly explain how unsupervised learning differs from supervised learning, focusing on their goals and data requirements.

Supervised and unsupervised learning represent two core approaches in machine learning, distinguished by their use of data and their primary objectives.

- Supervised Learning requires a labeled dataset where each example is associated with a known outcome. The goal is to infer a function that maps input features to these known labels, enabling the model to make accurate predictions on new, unlabeled data. The performance of a supervised model is directly measurable by comparing its predictions to the ground-truth labels using metrics like accuracy and precision. Focuses on prediction.

- Unsupervised Learning deals with unlabeled data, where no predefined outcomes are provided. Its purpose is exploratory: to uncover hidden patterns, intrinsic structures, or natural groupings within the data itself. Since there is no ground truth for direct comparison, the success of these methods is evaluated based on the quality and reasonableness of the discovered structure, often using internal metrics like the Silhouette Score for clustering. Focuses on Discovery.

Screenshots for Outputs

4.1 Load Titanic dataset and create `new_df` with required columns

```
survived    0
pclass      0
sex         0
age         0
dtype: int64
```

	pclass	survived	age	sibsp	parch	fare	body
count	1309.000000	1309.000000	1046.000000	1309.000000	1309.000000	1308.000000	121.000000
mean	2.294882	0.381971	29.881138	0.498854	0.385027	33.295479	160.809917
std	0.837836	0.486055	14.413493	1.041658	0.865560	51.758668	97.696922
min	1.000000	0.000000	0.170000	0.000000	0.000000	0.000000	1.000000
25%	2.000000	0.000000	21.000000	0.000000	0.000000	7.895800	72.000000
50%	3.000000	0.000000	28.000000	0.000000	0.000000	14.454200	155.000000
75%	3.000000	1.000000	39.000000	1.000000	0.000000	31.275000	256.000000
max	3.000000	1.000000	80.000000	8.000000	9.000000	512.329200	328.000000

4.2 Split into training and testing sets

- Training set: (732, 3)
- Testing set: (314, 3)

```
Training set: (732, 3)
Testing set: (314, 3)
```

4.3 Scale input features + train Decision Tree & Logistic Regression

- Models trained successfully!

```
Models trained successfully!
```

4.4 Evaluate the models

- **Decision Tree Performance:**
  - Accuracy: 0.7866, Precision: 0.8350, Recall: 0.6324
- **Logistic Regression Performance:**
  - Accuracy: 0.8025, Precision: 0.8033, Recall: 0.7206

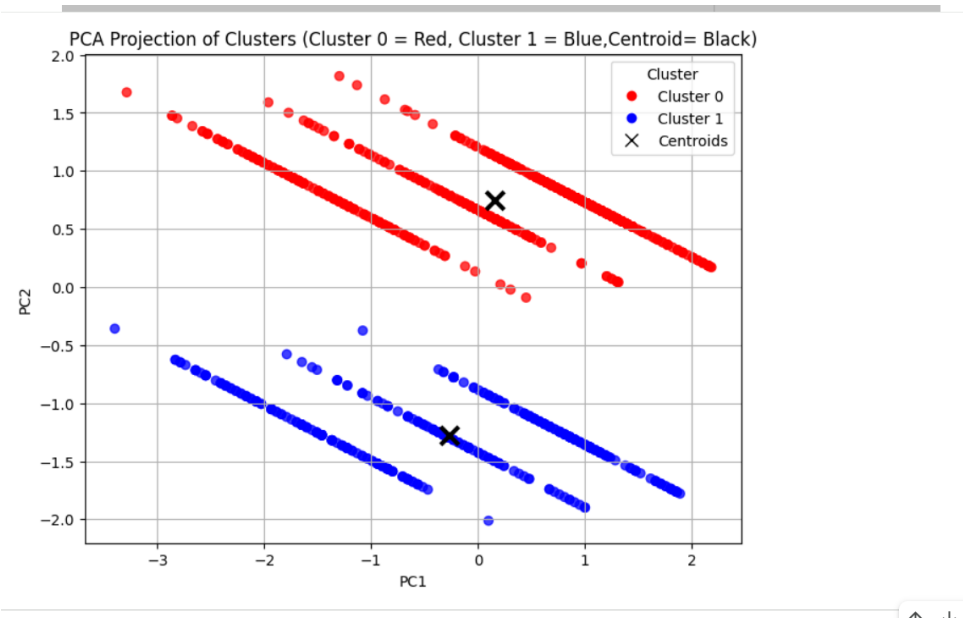
```
Decision Tree Performance:
Accuracy: 0.7866, Precision: 0.8350, Recall: 0.6324

Logistic Regression Performance:
Accuracy: 0.8025, Precision: 0.8033, Recall: 0.7206
```

5.1 Remove label + scale features + KMeans clustering

survived	0	1
cluster		
0	523	135
1	96	292

5.2 PCA visualization of clustering results



5.3 Silhouette score for cluster evaluation

- **Silhouette Score:** 0.3945
- Interpretation: Score near 1 is good; here it's moderate, meaning some overlap exists.

```
Silhouette Score: 0.3945
```

5.4 Cross-tab survival vs cluster

```
survived    0    1
cluster
0         523  135
1          96  292
```